Jamal Jennings Omosun

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Education

Carleton College *Northfield, MN*
Bachelor of Arts in Computer Science & Mathematics Sep 2022 – Jun 2026

- **Relevant Coursework:** Computability & Complexity, Data Structures, Software Design, Probability Theory, Linear Optimization, ODEs, PDEs, Real Analysis, Linear Algebra, Database Systems
- **Honors & Awards:** COMAP Mathematical Contest in Modeling (MCM) Honorable Mention, Eugster Endowed Student Research Fund, ColorStack Fellow, Caltech SURF Scholar, Caltech VURP Scholar

Technical Skills

Programming Languages	Python, R, Julia, Java, C, SQL (Postgres), JavaScript, Scheme
ML / Research Libraries	PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, Pandas, NumPy, pvlib, JuMP
Research Tools	Git, GitHub, LaTeX, Jupyter Notebooks, ArcGIS, OpenRefine, DeOldify, Voyant, Fusion360

Research Experience

California Institute of Technology *Pasadena, CA*
Undergraduate Researcher — AI Alignment & Large Language Models (SURF Scholar) Jun 2025 – Sep 2025

- Investigated model alignment and weight perturbation dynamics using open weights from Llama 3.1/4, probing neural network responses to politically-sensitive prompt matrices to identify systematic alignment failures.
- Implemented token-level activation engineering to selectively modify targeted network layer paths using PyTorch, TensorFlow, and Hugging Face Transformers, isolating causal pathways behind biased outputs.
- Quantified behavioral variance across neural clusters to derive mathematical bounds for ongoing algorithmic safety evaluation frameworks, contributing to reproducible metrics for alignment research.
- Documented experimental pipelines and delivered formal presentations of findings to departmental faculty and research stakeholders.

California Institute of Technology — Alvarez Lab *Pasadena, CA*
Undergraduate Research Assistant — Natural Language Processing (VURP Scholar) Jun 2023 – Aug 2023

- Co-led a computational social science project analyzing cryptocurrency misinformation mechanics on Twitter, processing a corpus of 300,000+ historical tweets under the mentorship of Dr. R. Michael Alvarez and Danny Ebanks.
- Built a Python text-processing pipeline automating tokenization, stop-word filtering, and high-dimensional vectorization, significantly reducing data cleaning overhead for the research team.
- Engineered and fine-tuned a multi-class TensorFlow text classification model to systematically segment and identify distinct misinformation clusters within the corpus.
- Authored a 10-page scientific research manuscript and delivered formal presentations at both Caltech and Carleton College.

Carleton College — FOCUS Program *Northfield, MN*
Student Quantitative Researcher — Sustainability Analytics Sep 2022 – May 2024

- Investigated institutional energy and infrastructure impact by integrating multi-source logistics data layers and tracking 3 months of empirical package freight information.
- Designed and distributed localized survey instruments collecting responses from 180 community members, applying data analysis to isolate 3 key institutional behavioral trends.
- Co-authored a research article published in the peer-reviewed Carleton FOCUS Magazine and presented empirical findings to 50 campus policy leaders, directly influencing structural workflow changes.

Academic & Technical Experience

Radian Generation *Remote*
Computational Engineering Extern — Solar Analytics Winter Fellowship Nov 2025 – Dec 2025

- Architected an automated Python data pipeline using Pandas, NumPy, and pvlib to ingest and clean multi-gigabyte meteorological and site production datasets.

- Trained an MLP Regressor within Scikit-Learn to model solar array performance degradation; implemented rolling time-series feature engineering on rainfall and energy output data to improve predictive accuracy.
- Exported automated reporting suites plotting model-to-actual variance calibration curves for stakeholder review.

Carleton College — Department of Computer Science *Northfield, MN*
Teaching Assistant & Academic Grader Sep 2025 – Present

- Facilitate technical instruction for 60+ undergraduates across Intro to Computer Science and Everyday Technologies, covering Python, algorithmic complexity, and hardware-software logic.
- Evaluate 30+ code submissions weekly, delivering rigorous feedback on code structure, optimization, and logical correctness; lead office hours focused on debugging and version control.

Carleton College — Digital Humanities Department *Northfield, MN*
Technical Teaching Assistant — Computational Methods Jan 2025 – Mar 2025

- Facilitated computational labs mentoring students in data-cleaning and spatial modeling frameworks including OpenRefine, Voyant, DeOldify, and ArcGIS; conducted troubleshooting clinics on 3D geometric modeling in Fusion360.

Selected Computational Projects

Solar Performance Soiling Analytics
Python, Scikit-Learn, pvlib, Pandas

- Integrated the Kimber physical model to simulate systematic array soiling ratios, leveraging rolling environmental statistics and an MLP Regressor to forecast energy yield loss from panel degradation across multi-year production histories.

Portfolio Optimizer
Python, Julia, JuMP

- Built a risk-aware asset optimization model using the Mean Absolute Deviation (MAD) framework and linear programming to enforce multi-asset diversification constraints against live market return histories.